

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

for

E-MALL

1. Introduction

1.1 Purpose :

This document is meant to delineate the features of E-mall, so as to serve as a guide to the developers on one hand and a software validation document for the prospective client on the other. The E-mall which is a web application is intended to provide complete solutions for small businesses as well as customers through a single get way using the internet. It will enable small businessmen to setup online shops, customer to browse through the shop and purchase them online without having to visit the shop physically. The administration module will enable a system administrator to approve and reject requests for new shops and maintain various lists of shop category.

1.2 Scope :

This system allows the customers to maintain their cart for adding or removing the product over the internet.

1.3 Definitions :

E-mall – Electronic mall

SRS- Software Requirement Specification

GUI- Graphical User Interface

Stakeholder- The person who will participate in system.

Ex. Customer, Administrator, Visitor etc.

1.4 References :

IEEE 830-1998 standard for writing SRS document.

Fundamentals of Software Engineering, 2nd ed. by Rajib Mall.

1.5 Overview :

This system provides an easy solution for customers to buy the product without going to the shop and also to shop owner to sell the product. This proposed system can be used by naïve users and it does not require any educational level, experience or technical expertise in computer field but it will be of good use if user has good knowledge of how to operate a computer.

2. Overall Description

The E-mall application enables small businesses to set up online shops, customers to browse through the shops, and a system administrator to approve and reject requests for new shops and maintain lists of shop categories. Also the developer is designing an online shopping site to manage the items in the shop and also help customers to purchase them online without visiting the shop physically. This online shopping system will use the internet as the sole method for selling goods to its consumers.

2.1 Product Perspective :

This product is aimed towards a person who doesn't want to visit the shop as he might not get time for that or might not be interested in visiting and dealing with lot of formalities.

2.2 Product Functions:

E-mall should support this use case

2.3 User characteristics :

User should be familiar with the terms like login, register, order system etc.

2.4 Principle Actors :

Two principle actors are Customer and Administrator.

2.5 General Constraints :

A full internet connection is required for E-mall.

2.6 Assumptions and Dependencies :

Working of E-mall needs Internet Connection.

3. Specific Requirements

3.1 Functional Requirements :

This section provides requirement overview of the system. Various functional modules that can be implemented by the system will be –

3.1.1 Registration

If customer wants to buy the product then he/she must be registered, unregistered user can't go to the shopping cart.

3.1.2 Login

Customer logs in to the system by entering valid user id and password for the shopping.

3.1.3 Changes to Cart

Changes to cart means the customer after login or registration can place order or cancel order of the product from the shopping cart.

3.1.4 Payment

In this system, we are dealing with the mode of payment by credit card, debit card, net banking etc.

3.1.5 Logout

After ordering or surfing for the products, customer has to logout.

3.1.6 Report Generation

After ordering the product, the system will send one copy of the bill to the customer's Email-address and another one for the system data base.

3.2 Non-Functional Requirements:

Following Non-Functional Requirements will be there in the insurance to the internet:

- (i) Secure access to consumer's confidential data.
- (ii) 24X7 availability.
- (iii) Better component design to get better performance at peak time.
- (iv) Flexible service based architecture will be highly desirable for future extension. Non-Functional Requirements define system properties and constraints.

Various other Non-Functional Requirements are:

- Security
- Reliability
- Maintainability
- Portability
- Extensibility
- Reusability
- Compatibility
- Resource Utilization

3.3 Performance Requirements:

In order to maintain an acceptable speed at maximum number of uploads allowed from a particular customer as any number of users can access to the system at any time. Also the connections to the servers will be based on the attributes of the user like his location and server will be working 24X7 times.

3.4 Technical Issues:

This system will work on client-server architecture. It will require an internet server which will be able to run PHP application. The system should support some commonly used browsers such as IE, mozilla firefox, chrome etc.

4. Interface Requirement

Various interfaces for the product could be-

- 1). Login Page
- 2). Registration Form
- 3). There will be a screen displaying information about product that the shop has.
- 4). If the customers select the buy button then another screen of shopping cart will be opened.
- 5). After ordering the product, the system will send one copy of the bill to the customer's Email-address.

Software Interface:

1. Operating System: Windows 7 Ultimate which supports networking.
2. JAVA development toolkit.

Hardware Interface:

Hardware requirements for insurance on internet will be same for both parties which are as follows:

Processor: Dual Core

RAM: 2 GB

Hard Disk: 320 GB

NIC: For each party

Communication Interfaces:

The two parties should be connected by LAN or WAN for the communication purpose.

5. System Design Specification

5.1 Architecture Design:

5.1.1 Data Flow Diagram(DFD):

It is a way of representing system requirements in graphical form; this led to modular design. A DFD describes a data flow(logical) rather than how they are processed. So they do not depend upon software, hardware, data structure or file organization. It is also known as 'bubble sort'.

A DFD is a structured analysis and a design tool that can be used for flowcharting in place of or in association with information-oriented and process-oriented system flowcharts.

A DFD is considered as an abstract of the logic of information-oriented or process-oriented system flowchart. The four basic symbols used to construct data flow diagrams are-